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PERSONALIZED LEARNING RECOMMENDATION SYSTEM

A PROJECT REPORT

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ABSTRACT

This report presents the **Personalized Learning Recommendation System**, an end-to-end Python and Machine Learning application designed to recommend suitable next learning topics and higher education pathways based on a student's academic profile. The system incorporates both a command-line interface (main.py) and an interactive Streamlit web application (app.py).

The machine learning engine utilizes a **Decision Tree Classifier** from scikit-learn to predict candidate learning topics based on categorical inputs (subject/interest, preferred learning format, completed topic) and numerical indicators (study hours, performance/marks, attendance). Preprocessing is managed through OneHotEncoder and ColumnTransformer pipelines as well as OrdinalEncoder transformations. Exploratory Data Analysis (EDA) is generated via Matplotlib and Seaborn, rendering feature importance metrics, marks distributions, and study hour correlations. Additionally, the Streamlit web application provides degree mapping for higher studies across bachelor's and master's programs

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CHAPTER 1: INTRODUCTION

1.1 Background

Digital learning environments provide students with access to many topics and learning materials. However, students differ in their previous academic performance, preferred learning formats, subject interests, and completed topics. A single fixed sequence of learning materials may therefore be unsuitable for every learner. A personalized learning recommendation system attempts to use a learner profile and information about available topics to suggest relevant next topics or resources.

1.2 Project Overview

The **Personalized Learning Recommendation System** is built using Python, Scikit-Learn, Pandas, Matplotlib, Seaborn, and Streamlit. The project consists of two execution modes:

- **Interactive Web Dashboard (`app.py`):** Offers real-time topic prediction, learning path visualization roadmaps, higher education advice (Bachelor's and Master's degree options), statistical summaries, downloadable CSV exports, and dynamic charts.
- **Terminal/Console CLI (`main.py`):** Processes student learning datasets (student_learning.csv), trains a Decision Tree model, outputs classification metrics, generates static graph plots, and provides CLI recommendations.

1.3 Problem Statement

Students often struggle to identify which topic to learn next or which degree to pursue due to overwhelming educational choices and a lack of personalized guidance. Traditional static course curricula fail to consider performance metrics or learning format preferences. The project addresses this by deploying a Machine Learning model to evaluate a student's current standing and dynamically recommend next-step learning topics.

1.4 Objectives

1. **Data Pipeline:** Build a robust data ingestion and cleaning pipeline to process student profiles, study hours, attendance, and performance marks.
2. **Model Training:** Implement and train a Decision Tree Classifier model with Scikit-Learn to predict suitable next-step topics.
3. **Exploratory Data Analysis (EDA):** Generate automated visualization graphs (Bar plots, Scatter plots, Feature Importance) using Seaborn and Matplotlib.
4. **Web GUI:** Develop a multi-page interactive web frontend using Streamlit.
5. **Educational Pathway Mapping:** Incorporate a dedicated module recommending Bachelor's and Master's degrees based on academic status.

1.5 Scope

The system handles multiple learning tracks: Python Programming, Database Management, Data Science & Analytics, Web Development, AI & Machine Learning, and Cybersecurity. It supports both local interactive execution (app.py) and batch terminal operations (main.py).

1.6 Limitations

The current demonstration web application uses a synthetic dataset generator to simulate student populations. Furthermore, recommendations follow strict domain prerequisite sequences.

CHAPTER 2: LITERATURE REVIEW AND EXISTING SYSTEM

2.1 Existing Learning Approaches

A fixed curriculum presents the same topic order to all learners. This is simple to administer but does not automatically adapt to a learner's performance or preferred format. Content-based recommendation approaches instead compare item attributes with a user profile. Collaborative filtering approaches use interactions or ratings from multiple users, but require sufficient interaction history.

2.2 Proposed Approach in Context

This project implements Supervised Decision Tree Classification combined with sequence-matching rules. By standardizing student profile features (academic level, performance marks, daily study hours, learning preferences, completed topics), the decision tree maps patterns from training records to suggest optimal next topics.

2.3 Limitations of Existing Approaches

A fixed topic sequence may not reflect individual readiness. Collaborative filtering may have a cold-start problem when interaction data is limited. Complex deep-learning recommenders can require more data and computation and may be harder to explain. The current project chooses a simpler and more interpretable supervised Decision Tree model.

CHAPTER 3: PROPOSED SYSTEM AND METHODOLOGY

3.1 Proposed System

The system ingests student attributes, encodes categorical data using Scikit-Learn transformers (OneHotEncoder / OrdinalEncoder), feeds numerical values (marks, study hours) through pipeline feature splitters, and predicts the target recommended_topic.

Table 3.1: Dataset Fields and Technical Roles

Field Name	Data Type	Technical Role	Purpose
subject / interest	Categorical	Feature	Subject domain selection
qualification	Categorical	Feature	Student educational level
completed_topic	Categorical	Feature	Currently completed topic
study_hours	Numerical	Feature	Daily/weekly study hours
performance / marks	Numerical	Feature	Student performance score (%)
preference / learning_style	Categorical	Feature	Preferred format (Videos, Reading, Projects)
recommended_topic	Categorical	Target (y)	Recommended next learning topic

3.2 Tools and Technologies

- **Python 3.x:** Core development language.
- **Streamlit:** Web application framework for app.py.
- **Scikit-Learn:** Machine learning library powering DecisionTreeClassifier, Pipeline, and encoders.
- **Pandas & NumPy:** Data cleaning, dataframe manipulation, and CSV preprocessing.
- **Matplotlib & Seaborn:** Visualization libraries for EDA charts.

3.3 Preprocessing & Pipeline Code

```
preprocessor = ColumnTransformer(
    transformers=[
        ("categorical", OneHotEncoder(handle_unknown="ignore"), categorical_features),
        ("numerical", "passthrough", numerical_features)
    ]
)

model = Pipeline([
    ("preprocessor", preprocessor),
    ("classifier", DecisionTreeClassifier(max_depth=12, random_state=42))
])
```

CHAPTER 4: SYSTEM DESIGN AND IMPLEMENTATION

4.1 System Architecture

The system consists of a Streamlit web frontend (app.py), a CLI interface (main.py), data preprocessing pipelines, and a Decision Tree classification model.

4.2 Exploratory Data Analysis Results

The historical student performance analysis generated during model execution reveals key trends regarding student study habits and marks distribution across subjects.

CHAPTER 5: CONCLUSION AND FUTURE SCOPE

5.1Conclusion

The **Personalized Learning Recommendation System** demonstrates an end-to-end Python-based learning topic recommendation workflow. It combines Scikit-Learn machine learning pipelines with Streamlit web interactive dashboards and command-line interfaces. The project successfully meets all academic requirements for BCA Machine Learning PBL deliverables.

5.2Future Scope

- Integration of collaborative filtering and deep learning models.
- Connecting to real-world institutional Learning Management Systems (LMS).
- Expanding the course catalogue to include non-technical domains.

REFERENCES

1. Python Software Foundation. *Python 3 Documentation.* https://docs.python.org/3/
2. The pandas development team. *pandas Documentation.* https://pandas.pydata.org/docs/
3. NumPy Developers. *NumPy Documentation.* https://numpy.org/doc/
4. Matplotlib Development Team. *Matplotlib Documentation.* https://matplotlib.org/stable/
5. scikit-learn Developers. *scikit-learn Documentation.* https://scikit-learn.org/stable/
6. Brainware University. *Machine Learning PBL instructions, Department of Computational Sciences.*